

Programming Fundamentals
Lab Mid Term Examination SPRING 2024

Total Marks: 40

Total Duration: 1 hour 30 minutes

Note:

- *There are a total of two questions.*
- *Attempt all the questions.*
- *Given sample input/output is for your understanding, your code must run on other input as well.*
- *Each question carries equal marks.*
- *Make use of as many functions (useful) as possible.*

Question 1: [CLO 1]

Marks: 20

Write a C++ program that reads a single line sentence in a C String from user and perform the following tasks on it using functions:

1. A void function that will replace all the lowercase vowels in the sentence with their corresponding upper-case vowel.
2. A function that returns the count of total number of spaces in the sentence.

Sample Run

Console Input:

Enter data: work hard for mid term

Console output:

After changing:

wOrk hArD fOr mId tErM

The no. of spaces is: 4

Question 2: [CLO 1]**Marks: 20**

PCB faced lose in HBL PSL 9 in business terms, as most of the people boycott PSL and did not go to stadium to watch matches, specially at National Stadium Karachi. The Manager of National Stadium wants you to write a program that calculate the total ticket sales after each match in Karachi. There are four categories of ticket- VIP, Premium, First Class and General.

After each game data is stored in file in the following format.

Data.txt

Category	ticketPrice	numberOfTicletsSold
VIP	600	19
Premium	300	75
First Class	200	46
General	100	240

Design a menu-based program to implement the following options. Perform the given tasks using functions. (Make as many functions as possible). User should press 'y' to continue and press 'n' for terminate

1. Search the category with the minimum number of tickets sold.
2. Display only those categories whose total sales does not exceed 15,000
3. Calculate total number of tickets sold.

Sample Run:**User Input:**

Please Select One of the following options:

1. Search the category with the minimum number of tickets sold.
2. Display only those categories whose total sales exceed 15,000
3. Calculate total number of tickets sold.

Enter your choice: **1**

Console output:

Category with the minimum tickets sold is: General

Do you want to continue (y/n) : **y**

Enter your choice: **2**

VIP has total sales: 11400

First Class has total sales: 9200

Do you want to continue (y/n) : **y**

Enter your choice: **3**

Total tickets sold: 380

Do you want to continue (y/n) : **n**